USB POWER 45W S8 / SINGLE BOARD, NO PER-RAIL EFUSES

S8 final: S7 converters kept; U6/U7 and the LM5067 negative switch removed; cans -> ceramics; two parallel 16-pin outputs; overvoltage guard on the fault latch; MAIN LED only (rail LEDs removed).

USB PD, main eFuse and VIN damper

RUN
VPD
VIN
MAIN_OK

CH224A: 20 V request
TPS16630: 4.186 A, the only switch
VIN damper 1R + 2 x 10u

File: 01_usb_input.kicad_sch

+12 V converter

MAIN_OK
VIN
+12V

TPS62933F
+12 V delivered rail

File: 02_positive_12v.kicad_sch

+5 V converter

MAIN_OK
VIN
+5V

TPS62933F
+5 V delivered rail

File: 03_positive_5v.kicad_sch

-12 V inverting converter

+12V
VIN
-12V

LMR38020F inverting buck-boost
-12 V delivered rail

File: 04_negative_converter.kicad_sch

Two output headers - monitors, overvoltage guard and MAIN LED

+12V
+5V
-12V
MAIN_LED_OK
RUN
V3V3
VPD
P12_OK
P5_OK
N12_OK
V20_OK
OV_N

Two Eurorack outputs J2 + J3
Monitors, OV guard
MAIN LED (only status LED)

File: 06_outputs_status.kicad_sch

FAULT POLICY
Each rail is monitored after its converter.
Persistent low rail (after ~0.57 s) or any overvoltage -> all OFF until RESET.
Main LED is not merely a +5 V lamp.

OUTPUTS: +12 / +5 / -12 on J2 and J3 in parallel
No per-rail current limit below the converter limits (TPS62933F 4.2-5.8 A peak; LMR38020F 2.6-3.8 A switch).
Targets 2.5 / 2.5 / 1.0 A; not a guaranteed 54.5 W simultaneous rating.

Always-on control - fault latch, overvoltage preset and RESET

MAIN_OK
N12_OK
OV_N
P12_OK
P5_OK
V20_OK
VPD
RUN
V3V3
MAIN_LED_OK

60 V always-on LDO / retained ARMED + FAULT
Manual RESET; OV_N presets FAULT

File: 05_fault_reset.kicad_sch

Design notes and pre-calculations: docs/s8-single-board.md
S8 final: single board, no per-rail eFuses, ceramic bulk, two output headers, OV guard, MAIN LED only

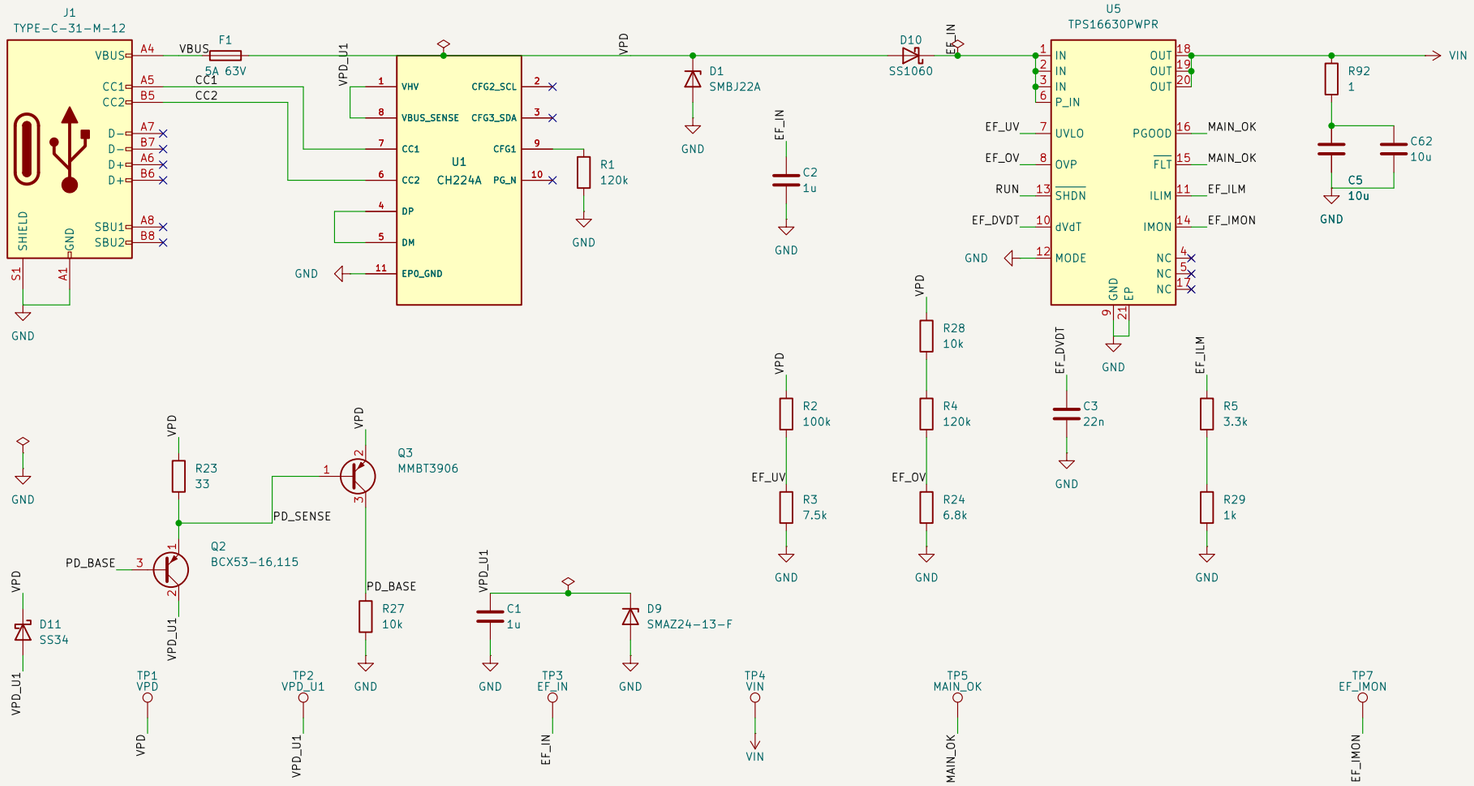
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Sheet: /
File: usb-power-45w-single.kicad_sch

Title: USB power S8 single board - wired power flow overview

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 1/7

USB PD, MAIN EFUSE AND VIN DAMPER



20 V PD request: U5 TPS16630 (60 V, 4.186 A nominal) is the ONLY switch: RUN low turns every rail off.
MAIN_OK = FLT + PGOOD at 3.3 V. R92 1R + C5/C62 2 x 10u damp the all-ceramic VIN bus (replaces the 47u can).

INTER-SHEET INTERFACE (see wired power-flow overview)

MAIN_OK
RUN
VIN
VPD

S8 single board, final schematic — see docs/s8-single-board.md

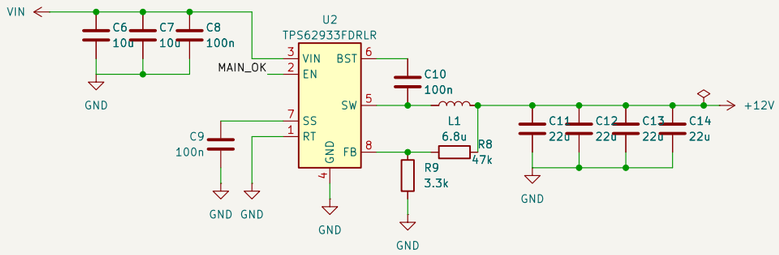
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Sheet: /USB PD, main eFuse and VIN damper/
File: 01_usb_input.kicad_sch

Title: USB PD, main eFuse and VIN damper

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 2/7

+12 V CONVERTER





TPS62933F +12 V, unchanged power stage; its output IS the delivered +12V (no rail eFuse).
Soft start C9 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C11–C14 here + C34 (J2) and C59 (J3) on page 06; the 100u can is gone.

INTER-SHEET INTERFACE (see wired power-flow overview)

+12V ◇—— +12V

MAIN_OK ◇—— MAIN_OK

VIN ◇—— VIN

S8 single board, final schematic — see docs/s8-single-board.md

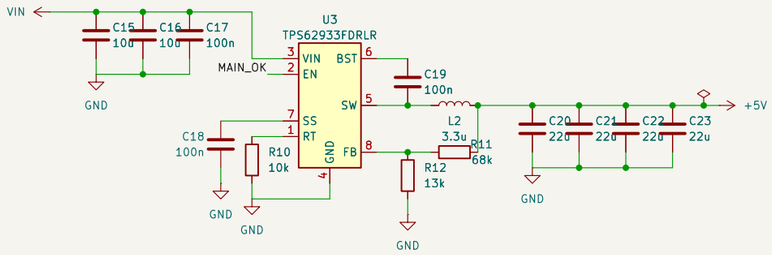
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Sheet: /+12 V converter/
File: 02_positive_12v.kicad_sch

Title: +12 V converter

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 3/7

+5 V CONVERTER





TPS62933F +5 V, unchanged power stage; its output IS the delivered +5V (no rail eFuse).
Soft start C18 100n = ~14.5 ms into rack capacitance. Own protection: 4.2–5.8 A peak limit, hiccup, OVP 112–118 %, TSD.
Rail ceramics: C20–C23 here + C35 (J2) and C60 (J3) on page 06; D21 TVS and the U20 OV guard on page 06.

INTER-SHEET INTERFACE (see wired power-flow overview)

+5V ——— +5V

MAIN_OK ——— MAIN_OK

VIN ——— VIN

S8 single board, final schematic — see docs/s8-single-board.md

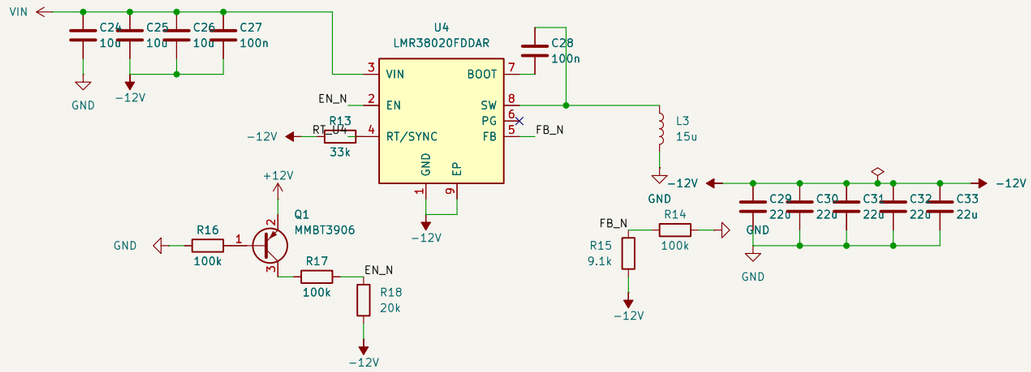
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Sheet: /+5 V converter/
File: 03_positive_5v.kicad_sch

Title: +5 V converter

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 4/7

−12 V INVERTING CONVERTER





LMR38020F inverting buck−boost; IC ground / exposed pad = the delivered −12V (no series switch).
4 ms soft start, 18 ms OCP blanking at start−up. Q1 follows +12V. C29−C33 here + C36 (J2) and C61 (J3).

INTER−SHEET INTERFACE (see wired power−flow overview)

+12V ◇ +12V

−12V ◇ −12V

VIN ◇ VIN

S8 single board, final schematic − see docs/s8−single−board.md

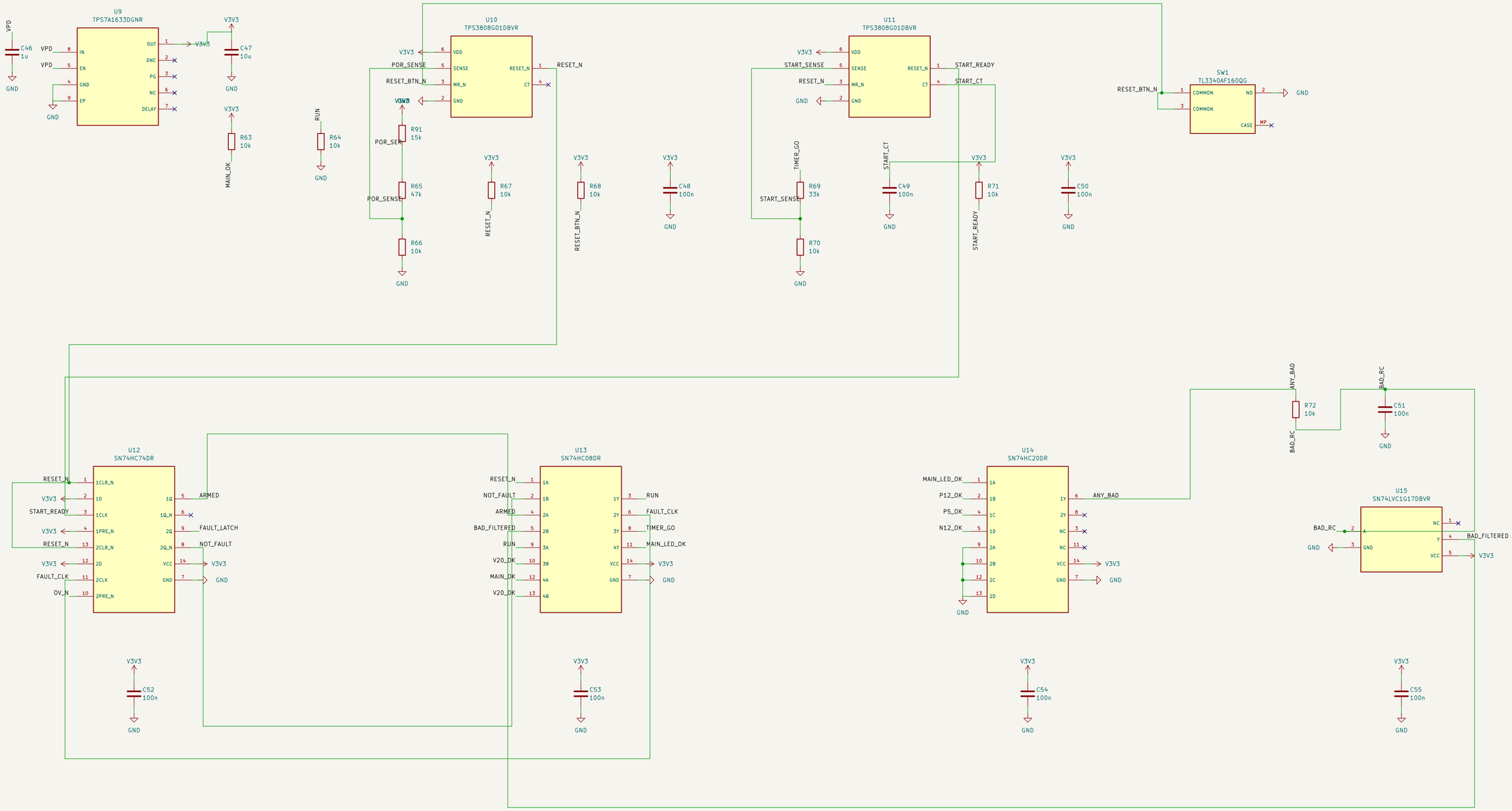
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Sheet: /−12 V inverting converter/
File: 04_negative_converter.kicad_sch

Title: −12 V inverting converter

Size: User	Date:	**Rev: S8 final**
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ALWAYS-ON CONTROL — FAULT LATCH, OVERVOLTAGE PRESET AND RESET



S8 single board, final schematic — see docs/s8-single-board.md

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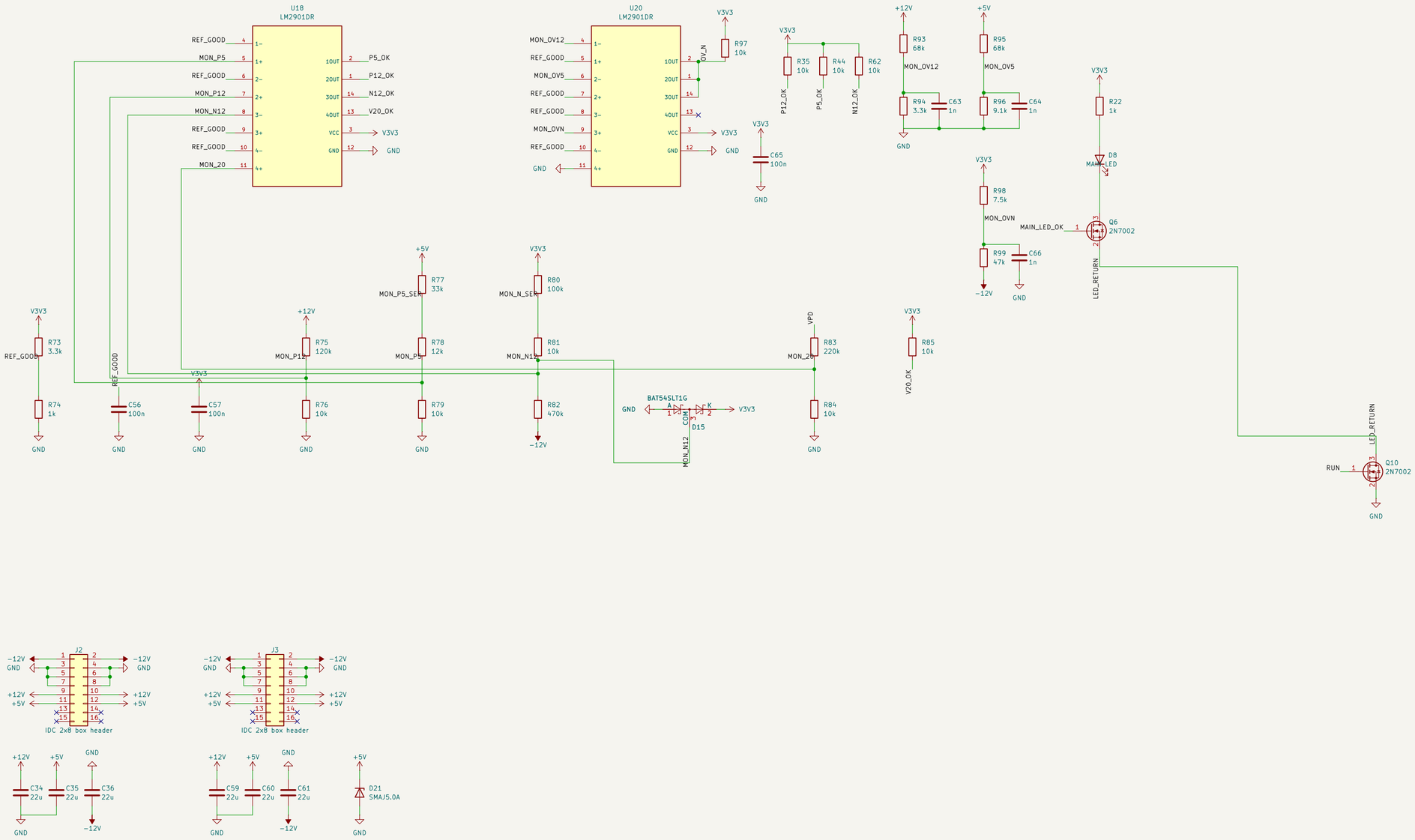
Sheet: /Always-on control — fault latch, overvoltage preset and RESET/
File: 05_fault_reset.kicad_sch

Title: Always-on control — fault latch, overvoltage preset and RESET

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 6/7

TWO OUTPUT HEADERS — MONITORS, OVERVOLTAGE GUARD AND MAIN LED

J2 + J3: two parallel 16-pin outputs, pin 1 = −12V. C34–C36 at J2, C59–C61 at J3 (22u each, replace the 100u cans).
D8 MAIN is the only status LED (rail LEDs removed); input-valid AND main-eFuse status AND RUN. Any bad rail after the
startup window or any overvoltage latches every rail off, so MAIN off = not working. P12/P5/N12_OK: logic + test pads only.
U20 overvoltage guard: +12V > 16.6 V, +5V > 6.5 V or −12V < −15.1 V nominal pulls OV_N → FAULT → U5 off, D21 clamps +5V transients.



INTER-SHEET INTERFACE (see wired power-flow overview)

+12V · +5V · −12V · MAIN_LED_OK · N12_OK
OV_N · P12_OK · P5_OK · RUN · V20_OK
V3V3 · VPD

S8 single board, final schematic — see docs/s8-single-board.md
Big Modular
Sheet: /Two output headers — monitors, overvoltage guard and MAIN LED/
File: 06_outputs_status.kicad_sch

Title: Two output headers — monitors, overvoltage guard and MAIN LED

Size: User	Date:	**Rev: S8 final**
KiCad E.D.A. 10.0.6		Id: 7/7